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Emil: An Emotion-Aware, Empathy-Driven Chatbot with Multi-Modal Emotion Detection

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Software

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Summary

Emil is an emotion-aware, empathy-driven chatbot that integrates multi-modal emotion detection with context-aware language generation. It processes **textual** and **visual** emotional cues in real time, combining facial expression recognition and natural language processing into a unified emotional state via a rule-based fusion strategy. This emotion, along with conversation history and relevant prior interactions, is injected into prompts for a locally hosted LLaMA3 model (via Ollama), enabling **privacy-preserving** (by deploying locally instead of using an online service API), **low-latency**, and **empathetic** responses.

Unlike traditional chatbots, Emil incorporates **user-personalized facial emotion classifiers** to adapt to individual expressiveness, as well as **Person-Centred Therapy (PCT) principles** in prompt engineering to enhance empathetic engagement.

Statement of need

Conversational AI systems often lack **emotional intelligence** and the ability to adapt responses based on nuanced human affect. While emotion recognition has been explored in isolation, few open-source projects provide an **integrated, privacy-preserving pipeline** for multi-modal emotion detection fused with real-time empathetic language generation.

Emil addresses this gap by: - Combining facial and textual emotion detection into a single, interpretable emotional state. - Supporting personalization via per-user facial emotion training. - Running locally to ensure **data privacy** and reduce dependency on cloud services. - Embedding therapeutic communication strategies to improve user emotional experience.

This makes Emil valuable for research on **mental health support tools**, **empathy-driven interfaces**, and **affective computing**.

Implementation

The Emil system is structured as an integrated pipeline with the following components:

1. **Input Capture** – Webcam video and user text are collected during typing events, ensuring naturalistic data gathering.
2. **Facial Emotion Recognition** – Real-time analysis using DeepFace for general emotion classification, combined with a custom-trained Support Vector Machine classifier for user-personalized accuracy, leveraging MediaPipe FaceMesh for feature extraction.
3. **Text Emotion Analysis** – NLP-based emotion detection using a fine-tuned DistilRoBERTa model (`j-hartmann/emotion-english-distilroberta-base`) from HuggingFace Transformers.

4. **Emotion Fusion** – A rule-based strategy prioritizing affective salience combines facial and textual cues into a single emotional state.
5. **Response Generation** – Prompt engineering integrates conversation history, retrieved similar past interactions, past and current emotions, and user input into a dynamic prompt for Ollama's local LLaMA3 model.
6. **Context Management** – Short-term session memory and long-term vector-store memory (SentenceTransformers) ensure continuity across conversations.
7. **User Interface** – A Tkinter-based GUI supports token-by-token streaming responses, live camera activation, and clear dialogue presentation.

The architecture is modular, allowing easy replacement or extension of emotion recognition, fusion, and response generation components.

Installation

```
# Clone the repository
git clone https://github.com/yulianakaraivanova05/empathetic-chatbot.git


# Install dependencies
pip install -r requirements.txt

# Run the application
python main.py
```

Usage

1. Launch the GUI application.
2. Type a message into the input field — the system will record short webcam snippets during typing for facial emotion detection.
3. Emil will generate an emotionally aligned response in real time, streaming tokens into the interface.

Personalised Classifier Training

Emil supports **user-specific emotion models** to improve facial emotion recognition accuracy. This feature allows the chatbot to adapt to individual expression patterns.

Step 1: Create a User Profile

Run the script below to collect facial expression samples:

```
python emotion/create_user_profile.py
```

- You will be prompted for a **username**.

- A dataset folder `user_data/<username>/` will be created with labelled images for each emotion.

Step 2: Train the Personalised Classifier

Train a Support Vector Machine (SVM) model on your dataset:

```
python emotion/train_user_emotion_model.py
```

- Enter the same **username** when prompted.

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- A trained model file (`<username>.pkl`) will be saved in the appropriate directory.

Step 3: Enable the Classifier in the Chatbot

To activate your personalised model:

1. Open `emotion/facial_emotion.py`.

2. Locate the `VideoRec` class.

3. Update the `user_profile` variable with your chosen **username**.

Once updated, Emil will load your personalised classifier alongside the default model during runtime.

Testing

Before running the facial detection tests, you need to provide 7 short clips for each emotion. These clips should be saved in tests/test_videos and named after each emotion. The tests script test_emotions.py has all the paths of the videos (e.g. happy.mp4 for happy)

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